

Telugu voice agent, from open source

A teardown of the Outpero Telugu calling agents, and a costed plan to build the equivalent yourself — at roughly a third of the running cost.

Researched **Aug 2026** Prices **INR, list** For **The Green Team**

PART 1

What Outpero actually is

Read their homepage and their Telugu page together and the shape is unambiguous. **They are an automation agency, not a voice-AI lab.** The homepage sells three "systems" — Revenue Capture from ₹60,000, Ops Efficiency from ₹1,00,000, Web Capture from ₹50,000 — plus 19 pre-built solutions from ₹14,999, delivered "live in 2 weeks." That's a services business with a fixed build kit.

The tell is on their own homepage: the Ops system lists *"Custom n8n/Make Logic"* as a deliverable. n8n is the glue. The voice layer is bought, not built.

WHAT EACH CLAIM MAPS TO

THEIR CLAIM	WHAT IT ACTUALLY IS
"10+ native Telugu voices" in a Voice Library	Cloned voices. Telugu voice actors recorded for 30–120 s, run through a cloning TTS. The names are their labels on cloned assets, not model names.
"3-second call" after a Facebook lead form	Meta Lead Ads webhook → n8n → outbound-call API. One automation node.
"Scale from 10 to 10,000 parallel calls"	The <i>telephony provider's</i> concurrency (Plivo/Exotel trunk channels), resold. Not their infrastructure.
"Transfer the live call to your team"	SIP REFER / warm transfer. A checkbox in Vapi, Retell, Bolna and LiveKit alike.
Voice Hub — recordings, transcripts, intent scores	A dashboard over webhook payloads. Recordings and transcripts come free from the platform; "intent score" is one LLM call after hangup.
"Telugu-first tuning... dialects, interruptions, Tanglish"	Prompt engineering, a good cloned voice, and endpointing tuned for Telugu. Real work — but config work, not model training.

Most likely stack underneath: a managed voice platform (Vapi / Retell / Bland, or Bolna given the India focus) + ElevenLabs or Sarvam for the cloned voices + GPT/Gemini for the brain + Plivo or Exotel for Indian PSTN + n8n for CRM glue + a custom dashboard. Their margin is the gap between ~₹4/min of vendor cost and what they bill. There is no moat here.

Why the demos sound ultra-realistic

Ranked by contribution. This ordering is the most useful thing in this document — it tells you where to spend your two weeks.

50% A cloned voice from a real Telugu speaker

Stock `te-IN` voices sound like a railway announcement. Sixty seconds of a real Hyderabad salesperson, cloned, sounds like a person — because it is one. This is the whole game; everything below is polish.

15% Native-script text normalization

`₹68.7 L` must reach the TTS as అరవై ఎనిమిది లక్షల డెబ్బై వేల. Every engine mangles digits, currency and initialisms in Indic scripts, and one mangled number ends the illusion instantly.

15% Under ~800 ms to first audio

Humans reply in 200–500 ms. Past a second of silence the caller says "hello?" and the spell breaks. An engineering budget, not a model choice.

10% Short turns

Cap the agent at two sentences / ~35 Telugu words. A twenty-second uninterrupted monologue is the single clearest giveaway of a bot.

5% Barge-in and backchannels

The caller must be able to cut the agent off mid-sentence and have it stop instantly. Sprinkle హా సర్, అవును, సరే while the LLM thinks.

5% The phone codec is helping them

PSTN is 8 kHz μ -law, which strips exactly the high frequencies where TTS artifacts live. The same voice that sounds slightly synthetic in a browser sounds human on a call.

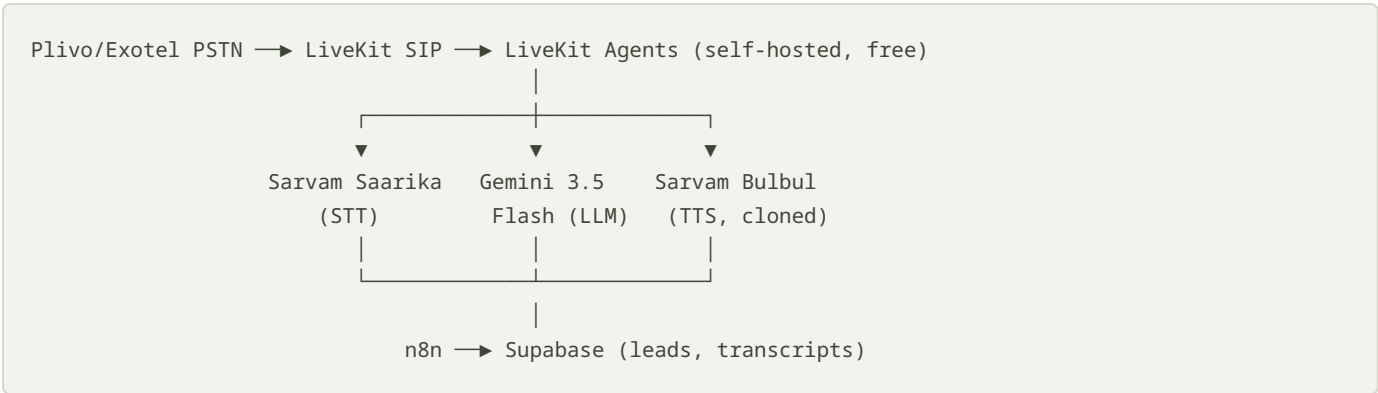
Their website demo players are almost certainly full-band studio renders. Never judge a voice on studio playback — A/B every candidate through an actual phone call before assuming they are ahead of what you can build in a week.

PART 2

Three stacks, priced

All per-minute maths assumes one minute of conversation: caller speaks ~25 s, agent speaks ~35 s $\approx$ 90 Telugu words $\approx$ **600 characters of TTS**, across ~5 turns. USD at ₹88.

Stack A — cascade on managed Indic APIs



LINE	ITEM	RATE	₹ / CALL-MIN
	Outbound PSTN (Plivo, mobile)	~₹0.71-0.80/min	0.75
	STT — Sarvam Saarika	₹30 / hour of audio	0.50
	LLM — Gemini 3.5 Flash	~\$1/M in, \$6/M out	0.15
	TTS — Sarvam Bulbul	₹30 / 10k chars × 600	1.80
	Orchestration VPS (~20 concurrent)	₹1,500/mo fixed	0.05
	Total	≈ \$0.037	3.25

TTS is 55% of the bill. That is the number to attack.

Stack B — speech-to-speech

Gemini Live native audio: no STT → LLM → TTS chain, ~300 ms response, Telugu supported, ~\$0.036/min ≈ ₹3.15, plus telephony ₹0.75 → ≈ **₹3.90/min**. Fewest moving parts and the best latency — but **you cannot use your own cloned voice**, and the cloned voice is half the realism. Wrong trade for a sales agent trying to sound like a specific human; right choice for an inbound receptionist where neutral-but-fast wins. Build it in week 1 as your latency baseline; it takes an afternoon.

Stack C — self-hosted STT + TTS

LINE	ITEM	₹ / CALL-MIN
	Outbound PSTN	0.75
	LLM (Gemini Flash; ~0 if self-hosted)	0.15
	One L4 24 GB @ \$0.39/hr ≈ ₹34/hr, STT + TTS, ~10 concurrent	0.06
	Orchestration VPS	0.05
	Total ≈ \$0.012	1.00

Break-even. Self-hosting saves ~₹2.25/min. A GPU running 24/7 costs ~₹816/day. You need **~360 call-minutes/day (6 hours of talk time)** to justify it.

Build Stack A, instrument it, migrate to Stack C past ~350 call-min/day. Starting at C means three weeks on GPU autoscaling instead of on the conversation — which is where all the quality actually lives.

PART 3

Components, with repos

ORCHESTRATION — PICK ONE

REPO	LICENSE	VERDICT
livekit/agents ← pick	Apache-2.0	WebRTC rooms, native SIP and phone numbers built in — no Twilio bridge. Best turn-detection model ships in-repo. Runs OpenAI's and xAI's production voice. Free self-hosted.
pipecat-ai/pipecat	BSD-2	Pipeline-of-processors, largest plugin library, near-daily commits. Choose it if you want to hand-tune every stage — e.g. a custom Telugu normalizer between LLM and TTS.
bolna-ai/bolna	MIT	India-native: Plivo/Exotel/Vobiz + BYO-SIP, agents as JSON, Sarvam integrated. Fastest path to a ringing Indian phone. Caveat — the README currently asks for maintainers. Borrow the telephony servers; don't make it your core.
TEN-framework	Apache-2.0	Strong real-time multimodal, C++/Go core. Heavier to learn. Only if you need avatar/video later.

Framework choice barely affects latency — that's set by your model stack (cascade adds 300–800 ms, speech-to-speech ~300 ms; framework overhead is tens of ms). Choose on telephony and ecosystem.

TTS — WHERE THE REALISM LIVES

MODEL	LICENSE	VERDICT
ai4bharat/IndicF5	open	1,417 h of Indian speech, 11 languages incl. Telugu. Reference-audio cloning — clip + its transcript. The default open choice. Not natively streaming; chunk by sentence.
praxelhq/praxy	Apache-2.0	Chatterbox + Telugu LoRA, with a published benchmark beating Sarvam Bulbul on Telugu retroflex collapse (26.7% vs 33.3%) and a router that transliterates Tanglish before synthesis — exactly the problem Outpero advertises. Brand new (Apr 2026), zero stars, unproven. Evaluate; don't bet on it yet.
resemble-ai/chatbox	MIT	350M params, sub-200 ms tuned, emotion control, zero-shot cloning. 65% blind preference over ElevenLabs (English). ~8 GB VRAM. Good fine-tune base.
canopyai/Orpheus-TTS	Apache-2.0	Llama-based, true streaming, guided emotion, 150M–3B. Needs a Telugu fine-tune.
Sarvam Bulbul (API)	commercial	₹30/10k chars, Indic-tuned, no GPU. The pragmatic Stack-A default.
ElevenLabs Multilingual v2	commercial	Best cloning, 5–10× Sarvam's price. Use it to prove the voice, then replicate on IndicF5.

SPEECH TO TEXT

MODEL	NOTES
AI4Bharat IndicConformer 600M	22 Indian languages incl. Telugu. The strongest open Indic ASR. Streaming-capable.
faster-whisper + vasista22/whisper-telugu-large-v2	Easiest path. Needs chunking for streaming, heavier GPU.
collabora/WhisperLive	Ready-made streaming wrapper if you go Whisper.
Sarvam Saarika (API)	₹30/hour, Indic-tuned, zero ops. Stack-A default.

TURN-TAKING – DO NOT SKIP THIS

- [snakers4/silero-vad](#) — the baseline VAD everyone uses. MIT, tiny, CPU.
- **LiveKit turn detector** (in [livekit/agents](#)) — semantic end-of-turn detection; knows "నేను అనుకుంటున్నాను..." isn't finished. Cuts false interrupts dramatically.
- **Smart Turn** (in [pipecat](#)) — the equivalent on that side.

Endpointing is the difference between an agent that talks over people and one that doesn't. Budget real tuning time — it's the most under-rated component in every voice stack.

TELEPHONY (INDIA)

PROVIDER	OUTBOUND MOBILE	NOTES
Plivo	~₹0.71/min	Best API and docs, 100 default concurrency, first-class in Bolna and LiveKit SIP.
Exotel	₹0.80–1.20	Most India-native; strongest on DLT paperwork.
Direct SIP (Tata/Airtel)	₹0.60–1.20	Cheapest at volume, most paperwork. Worth it past ~50k min/mo.
Twilio India	highest	Don't, for domestic India.

Inbound DID ₹0.40–0.90/min. Toll-free inbound ₹1.20–2.50/min.

EVERYTHING ELSE

- **Glue:** [n8n-io/n8n](#), self-hosted — the same tool Outpero uses. Meta Lead Ads → webhook → trigger call.
- **Data + dashboard:** Supabase for leads, transcripts and recordings; [Metabase](#) or a Next.js page as your Voice Hub.
- **Post-call intelligence:** one Gemini Flash call over the transcript → intent score, budget, objection tags, next action. ~₹0.10/call.

PART 4

The realism checklist

Ship none of these and you have a robot. Ship all of them and you're at parity.

1. **Clone a real voice.** Hire a Telugu voice actor for two hours. Record 10–15 minutes clean — no room echo, no compression, 48 kHz — reading the objection blocks from the cold-call script, so you capture the *sales register*, not neutral narration. Get a signed release covering synthetic reproduction. This one step outranks every model choice.
2. **Normalize text before TTS.** Numbers, currency, dates and initialisms into Telugu words. Praxy ships `indic_numbers.py` for exactly this — steal it. Never send ₹68.7L or AQI 12 raw.
3. **Cap turn length in the system prompt.** "At most 2 sentences, under 35 Telugu words, never more than 3 items." Enforce it — truncate server-side when the LLM overruns.
4. **Stream TTS sentence-by-sentence.** Start speaking on sentence one while the rest generates. Halves perceived latency for free.
5. **Hold sub-800 ms to first audio.** Typical budget: VAD close 250 ms → STT final 150 ms → LLM first token 250 ms → TTS first chunk 150 ms. Instrument each stage separately.
6. **Barge-in that actually stops.** On detected speech, kill the audio buffer immediately — not at the end of the current chunk.
7. **Filler while thinking.** If the LLM hasn't returned in 400 ms, play a pre-rendered "హ్ సర్...". Pre-render once; never synthesize live.
8. **Warm human transfer.** SIP REFER on "మనిషితో మాట్లాడాలి". Test it on day one — it's the most business-critical path in the system.
9. **Handle Tenglish input.** Real callers say "sir, price entha?" Your STT must not drop the English, and your prompt must not correct the caller's register.
0. **Tune on 8 kHz.** Evaluate every voice through a real PSTN call, never laptop speakers.
1. **Re-listen weekly.** Pull 20 real calls, mark every moment you'd have hung up, fix the top three. This loop is what separates a demo from a product — and it's the only part nobody can sell you.
2. **Disclose the AI in the first sentence.** Compliance, yes — but also, callers who feel deceived hang up harder and complain louder.

PART 5

India compliance, non-negotiable

Outbound commercial calling in India in 2026 is regulated and actively enforced — TRAI's own detection models look for synthetic-voice signatures and abnormal call patterns. Over 47,000 numbers were disconnected in Q1 2026 alone, including legitimate businesses with sloppy DLT registration. Penalties run to ₹10 lakh.

Before your first live dial:

- ☐ **Register as a telemarketer on the DLT platform** — entity, sender IDs, and your call scripts.

- ☐ **Use the correct number series** — 140 promotional, 1600 transactional. Promotional traffic from a normal 10-digit mobile is the fastest route to disconnection.

- ☐ **Scrub every list against DND** before each campaign, not once.

- ☐ **Capture revocable consent** with timestamp and source. A Meta Lead Ad submission is consent for *that* enquiry — log the lead ID against the call record.

- ☐ **Call only 09:00–21:00** local time.

- ☐ **Disclose the AI** in the opening line and label synthetic voice per the 2026 IT Rules amendment. Never impersonate a named real person.

- ☐ **Honour opt-out instantly and permanently** — wire it to a hard suppression list, not a CRM tag.

- ☐ **DPDP Act:** recordings and transcripts are personal data. Define retention, restrict access, support deletion on request.

The cheapest stack in the world is worthless if the numbers get cut off in week three.

PART 6

Fourteen days

DAYS 1–2 **Voice asset**

Book the voice actor. Record 15 minutes. Clone into ElevenLabs *and* IndicF5 from the same source clip, A/B them on the 17-clip test set — through a phone call, not speakers. Pick a winner.

DAYS 3–4 **Local loop**

LiveKit Agents locally, browser mic, no telephony: Saarika → Gemini Flash → your voice. Get one full turn working. Spend an afternoon on Gemini Live speech-to-speech as the latency baseline to beat.

DAYS 5–6 **Conversation design**

Port the nine blocks and seven objection handles into the system prompt as a state machine: greet → qualify → pitch → objection → close → book/transfer. Add the turn cap and the Telugu normalizer. Give this the *most* time — it's where the quality is.

DAYS 7–8 **Telephony**

Plivo trunk → LiveKit SIP, outbound and inbound. Test warm transfer. Start DLT registration now — approval takes days and it will be your critical path.

DAYS 9–10 Latency and endpointing

Instrument the four-stage budget. Add turn detection, barge-in, pre-rendered fillers. Target sub-800 ms. This is the gap between "impressive" and "real".

DAYS 11–12 Data loop

n8n: Meta Lead Ad → Supabase → trigger call → post-call summary → back to Supabase. Build the minimal Voice Hub view: calls, duration, outcome, transcript, recording, intent score.

DAYS 13–14 Pilot and regression set

Fifty real calls to consented leads. Listen to all fifty. Build a fixed 20-case Telugu regression suite from the failures and re-run it on every prompt change, forever.

Then, and only then: if you're past ~350 call-min/day, migrate STT and TTS onto an L4 and take the bill from ₹3.25 to ₹1.00 per minute.

Bottom line

Outpero is a well-executed assembly job on bought parts, sold at agency prices. Nothing on that page requires technology you cannot self-host this month.

Your real costs: **~₹3.25/min managed, ~₹1.00/min self-hosted**, plus one voice actor (₹5–15k, one-time) and about two weeks of build time. What decides whether it sounds ultra-realistic is not the framework — it's the quality of that one voice recording, whether your numbers are spelled out in Telugu, and whether you keep the agent's turns short.

Everything else is plumbing.

Sources. Frameworks — [Soniox](#), [ThinnestAI](#), [Evalgent](#). Models — [IndicF5](#), [Praxy](#), [Bolna](#), [MarkTechPost ASR 2026](#), [Speakeasy TTS 2026](#). Pricing — [Sarvam](#), [Plivo India](#), [Runpod](#), [Caller Digital telephony comparison](#). Compliance — [TRAI DLT for AI calling](#), [AI cold calling India 2026](#), [DPDP + TRAI + RBI guide](#).

Figures researched August 2026 and approximate. Verify vendor pricing and DLT requirements before committing.